

Indoor Farming

- Indoor farming is known as **vertical, horizontal, controlled environment, or urban farming**. This is a new technique of farming in which maximum benefit is taken from the land.
- In this farming, success is achieved through **a controlled environment inside the building structure, such as a greenhouse, warehouse, basement, or unused industrial building**.
- Large equipment **like tractors or combines** is not used in indoor farming. The need for water, fertilisers, and other inputs is **also less as compared to traditional farming**. Doing this farming on a **large scale is more profitable**. New equipment like **cameras and software** is used in this farming. In this farming, **we grow onions, green leafy vegetables, microgreens, potatoes, tomatoes, etc.**

Types of indoor farming businesses:

1. **Hydroponic Farming:** The word hydroponic is derived from the **Greek language**. It is made up of two words: "**hydro,**" which means "**water,**" and "**ponein,**" which means "**to labour, work, or toil.**" In this technique of agriculture, soil is not required for the cultivation of plants. The nutrients required for the growth of plants are mixed in water to make a solution and supplied through **rockwool**. Through this farming, farmers are able to provide fresh food **365 days a year**.
2. **Aeroponic farming:** This is a technique of growing plants without soil. **The roots of the plant are hung in the air and irrigated with a solution of nutrient-rich water through mist**. In hydroponics, the roots remain **immersed in water**; hence, it is completely different from it.
3. **Aquaponics Farming:** Aquaponics is a method in which **fish as well as plants are grown without soil**. Fish waste contains beneficial bacteria, which convert it into essential nutrients for the plants, and the plants filter and purify the wastewater to create a proper environment for the fish to live. This method consumes less water and fertilisers than traditional farming.
4. **Vertical Farming:** It is a type of farming technique in which farmers **grow plants in vertical, stacked layers indoors without soil**. **Green leafy vegetables** are usually grown in this farming. Plants are grown through a controlled environment **such as light,**

temperature, humidity, and nutrients. You will be surprised to know that this farming uses **98% less water, 90% less light energy, and up to 85% less nutrients than traditional farming.**

- 5. Controlled environment farming:** In this technique, crops are grown by creating optimal conditions by carefully controlling the proper environment, **such as temperature, humidity, light, and CO2 levels.** Farmers grow crops in protected structures such as **greenhouses.**
- 6. Container farming:** It is a method of growing crops in shipping containers using climate control and artificial light. It is a form of vertical farming.

Advantages of Indoor Farming:

- Land is used efficiently.
- It provides the ability to produce seasonal food grains throughout the year.
- Production is not affected due to the absence of adverse conditions.
- Input costs are lower than traditional farming.
- It helps in making agriculture a sustainable practice.

Challenges in indoor farming:

- **Initial cost: Marketing, lighting, electricity, and growing equipment cost a lot to start indoor farming. Loan repayment and land rent are also very high.**
- **Lighting Complexity:** Indoor farming has been expanding rapidly over the past several years due to the falling prices of **light-emitting diodes (LED).** Both **LED lights and fluorescent lights** give the same amount of light energy. But LED lights require half the amount of energy compared to fluorescent lights. Research was conducted by the Philips company in the year 2015, which found **that plants respond differently to different spectrums of light and colour at different points of their growth cycle. LED lights are unable to mimic sunlight.**
- **Skilled workers:** Indoor farming requires skilled and experienced workers to monitor the crops. Farmers face difficulties in finding skilled workers.
- **Limited crop variety:** The crop range of indoor farming is limited as compared to traditional farming. In the field, we can grow many types of crops like rice, soybean, wheat, corn, etc., but in indoor farming we can only **grow green leafy vegetables,**

herbs, and small fruits. While chillies, tomatoes, and cucumbers are grown in greenhouses.

- **Consumer acceptance and market awareness:** Vegetables grown in indoor farming are less preferred by consumers. They still prefer vegetables grown using traditional methods. **Certification and branding** are necessary to attract their attention so that they can trust the products of indoor farming.

Market potential and growth opportunities in India:

1. Market potential:

- **Global:** The global indoor farming market size was **US\$40.51 billion in 2023**. It is projected to reach **US\$118.17 billion by 2033**, growing at a compound annual rate (CAGR) of **11.3%**.
- **In India:** The market size was **US\$0.823 billion in 2022**; it is projected to reach **US\$1.740 billion by 2029**, growing at a compound annual rate (CAGR) of **11.29%**.

2. Growth opportunities:

- Indoor farming is a sustainable practice that is being **moulded as advanced agriculture through AI**. Its growth can be seen day by day in urban areas. People are becoming conscious about their health. Indoor farming gives many times more production in less space than flat land. This farming requires less water and fertiliser.

Conclusion:

Indoor farming is a sustainable farming method. Indoor farming is the solution to challenges like **land scarcity, food security, and climate change**, which is revolutionising the fields of agriculture. Farmers can earn good money by starting their own startup with high investment and government support.

Sources:

<https://bowery.co/indoor-farming/>

<https://viemose-dgs.dk/indoor-farming/>

<https://www.lettusgrow.com/aeroponic-technology>

<https://url-shortener.me/J7H>

<https://url-shortener.me/J7I>

<https://www.lettusgrow.com/blog/what-is-a-container-farm>

<https://url-shortener.me/J7I>

<https://dirt-to-dinner.com/the-challenges-of-indoor-farming/>

<https://url-shortener.me/J7J>

<https://url-shortener.me/J7K>